Queen Mary University of London
School of Engineering and Materials Science

EMS520U
Neuromechanics and Bioelectricity of Movement

EMG Lab Report

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Date of attendance of the activities in the lab: 27th February 2024.

ABSTRACT

The aim of this experiment was to study how muscle activation differs between the triceps and biceps during bicep curls, using electromyography (EMG) and considering factors like weight, gender and individual disparities like muscle architecture. An experiment was conducted to assess this objective using different weights and two subjects of different genders. Surface EMG electrodes were placed on the triceps and biceps of the two test subjects, and they alternated in performing two sets of bicep curls with each weight. MATLAB was employed to analyse the raw data by computing maximum amplitude, mean rectified value, the root mean square and other statistical features of the EMG signals. The results showed that there was higher muscle activation in the biceps compared to the triceps during the lifting phase of the curl, consistent with the biceps' role as the primary mover during the bicep curl exercise. Noise interference was mitigated by ensuring proper electrode placement and minimizing external sources of interference, to prevent noisy EMG signals. Further analysis revealed distinct frequency patterns for the biceps and triceps during the exercise.

INTRODUCTION

Electromyography (EMG) serves as a valuable tool assessing muscle activation during various physical activities. This study focuses specifically on how muscle activation differs between triceps and biceps when performing a bicep curl.
The aim of this experiment was to record and analyse the electrical signals generated by these muscles throughout the curling motion.
As participants engaged in biceps curls with varying weights, we expected to observe notable differences in EMG signals. The hypothesis guiding this study suggests that the triceps and biceps will exhibit different activation patterns. The experimental setup involved careful consideration of factors that could potentially affect the results of this experiment such as weight, gender and individual variability by including the use of different weights and different participants.
Surface EMGs were used due to their non-invasive nature and low signal to noise ratio. Motor neurons are nerve cells extending beyond the central nervous system, transmitting action potentials to skeletal muscles for movement. Each motor neuron connects with muscle fibres at neuro-muscular junctions, forming a functional unit known as a motor unit [1]. Electromyography (EMG) studies muscle function by interpreting electrical potentials during contraction or relaxation. Motor Unit Action Potential (MUAP) represents combined muscle fibre action potentials within a motor unit [2], showing variability due to superposition of individual action potentials of each muscle fibre. A sequence of MUAPs is termed Motor Unit Action Potential Train (MUAPT).
The subsequent sections of this lab report will detail the experimental protocols, findings, and thorough analysis of the EMG data.

Materials and Methods

Experimental Setup:

This section outlines the materials and methods used to record and analyse EMG signals in this experiment. The Delsys Trigno wireless EMG system served as the primary apparatus used for recording the EMG signals. This system included a built-in analysis toolbox, which allowed monitoring of EMG signals in real time.
Two distinct subjects, one male and one female of similar age, participated in the experiment. Both subjects used their dominant hands for the bicep curls.
Sensors 4 and 7 respectively were placed on the triceps and biceps of Subject 1, and Sensors 8 and 9 respectively were placed on the biceps and triceps of subject 2. The sensors were positioned to align the sensor's arrow with the direction of the muscle fibres. The arms of each subject were sanitised and cleaned before sticking the EMG sensors on, to ensure optimal skin contact and prevent noise production in the EMG signal.
Bicep curls were performed with different weights – no weight, 2kg, 4kg, and 6kg. Ten repetitions were performed for each weight, and a metronome set at 60 beats per minute was used to ensure consistency in the timing, with one curl performed every three seconds. Each subject performed two sets for each weight and took alternating turns to allow sufficient rest time.

Data Processing and Analysis:
The raw data was recorded for each bicep curl repeat for each subject and converted into an excel file to be processed.
The raw EMG data recorded during the bicep curls underwent processing using MATLAB. Notch and bandpass filters were applied, followed by rectification and smoothing of the signal. To estimate the EMG sample rate for the dataset, the timestamps provided in the data table were used. By calculating the time difference between two consecutive timestamps, the sample time (T) was derived, and the sample rate (f) was calculated as 1/T. Statistical analyses were also performed using MATLAB to extract relevant features from the EMG data. The maximum amplitude of each signal, signal rectification to detect the start of muscle activity, the mean value of the rectified signal before, during, and after contraction, and the root mean square of each signal at different stages of the bicep curl were computed.

RESULTS

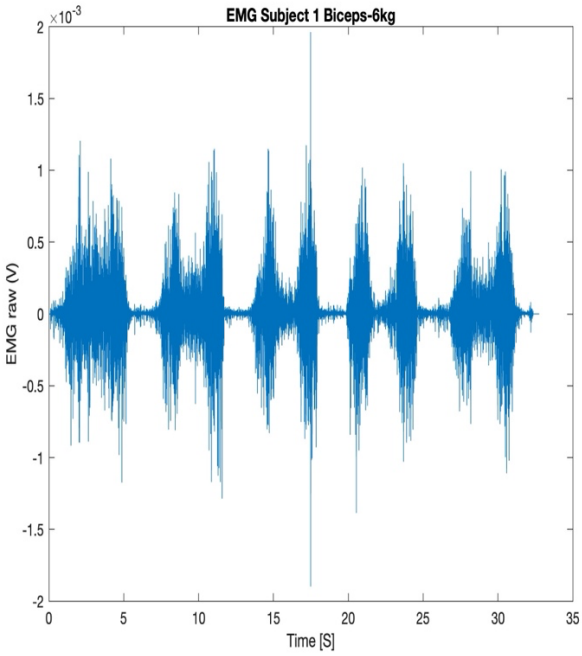


Figure 1- Raw data visualisation for Subject 1 biceps-6kg

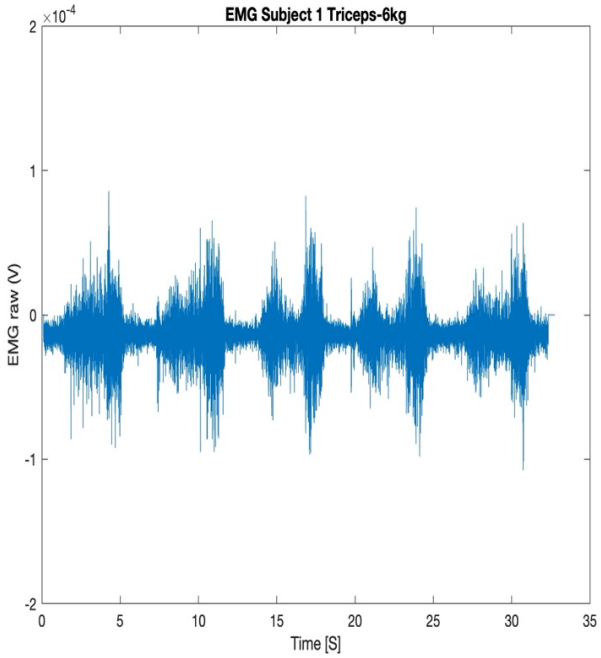


Figure 2- Raw data visualisation for subject 1 triceps-6kg

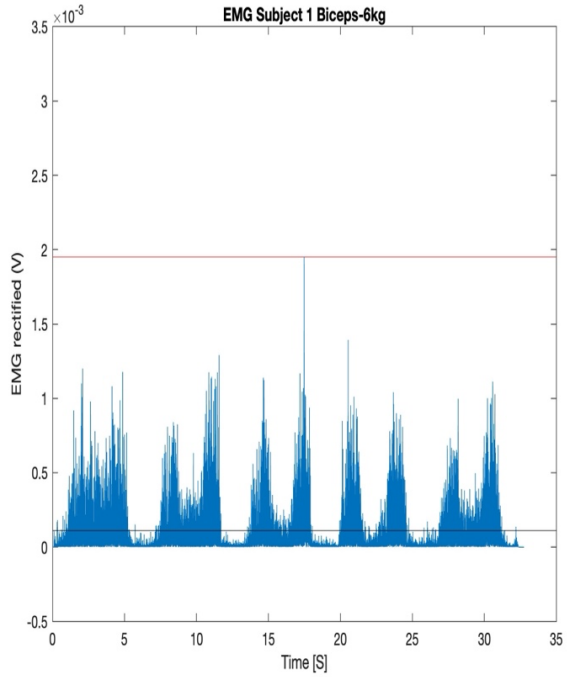


Figure 3- Rectified data for Subject 1 biceps-6kg

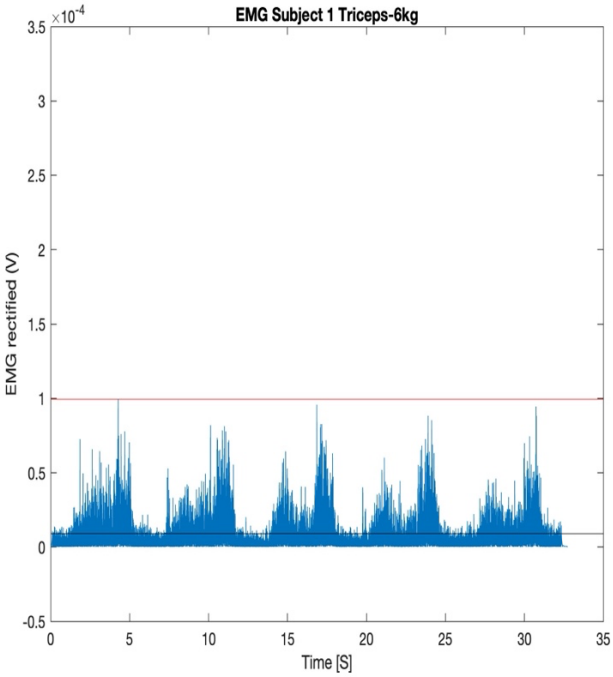


Figure 4-Rectified data for Subject 1 triceps-6kg

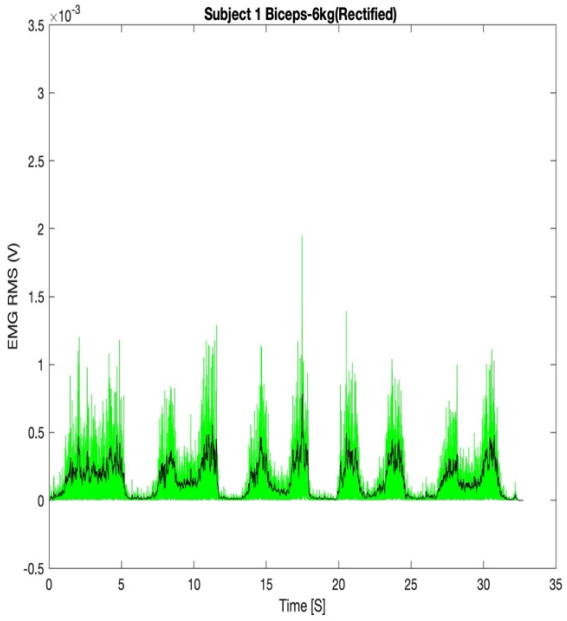


Figure 5- Root mean square for Subject 1 biceps-6kg

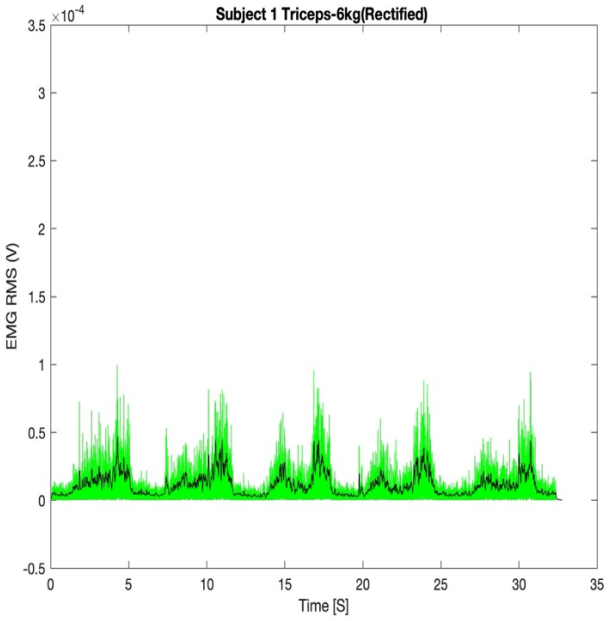


Figure 6- Root mean square for Subject 1 triceps-6kg

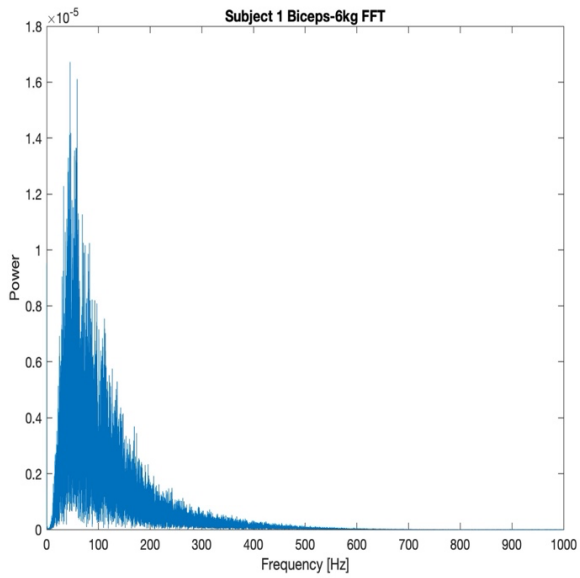


Figure 7- FFT for Subject 1 biceps-6kg

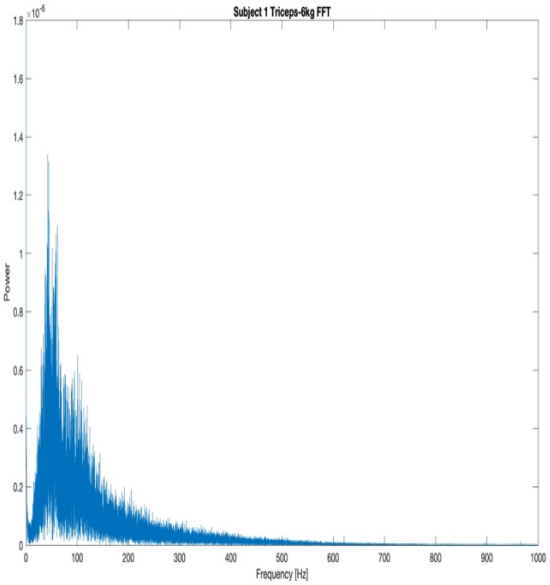


Figure 8- FFT for Subject 1 triceps-6kg

DISCUSSION

This experiment aimed to assess the difference in muscle activation between the triceps and biceps during bicep curls.

Firstly, it was expected that during the bicep curls, both muscles would undergo greater muscle activation as the weight increased, resulting in higher EMG signal amplitudes for both muscles.

Figures 1 and 2 depict the raw EMG signals of the triceps and biceps in the dominant arm of Subject 1 while performing the bicep curls with a 6kg weight. Comparison between the muscles in these graphs reveals a greater amplitude in the biceps as compared to the triceps, confirming that there is indeed a significant difference in muscle activation between the two muscles during bicep curls. Evidently, there is higher activity in the biceps than the triceps during the curl bicep curl.

The biceps are the primary movers in the bicep curl exercise, responsible for flexing the elbow joint [3]. Therefore, it was expected that the biceps would show a higher overall EMG signal amplitude compared to the triceps during this experiment.

During the lifting (concentric) phase of the bicep curl, the biceps undergo significant activation to generate force and lift the weight [4], resulting in higher maximum amplitude values for the biceps compared to the triceps, reflecting the greater intensity of muscle activation. Similarly, the mean rectified values, which provide an indication of the average signal strength over time, were also higher for the biceps during this phase.

On the contrary, during the lowering (eccentric) phase of the curl, the triceps engage to control the descent of the weight. While the triceps still undergo some activation during this phase, it is evidently less intense as compared to the biceps during the lifting phase.

The triceps therefore produced lower EMG signal amplitudes during the curling phase but higher amplitudes during the lowering phase of the movement. This is depicted clearly in Figure 4 as the amplitudes of the signal produced by the triceps decreases with time, with the maximum amplitude being produced at the onset of the exercise as the arm is in its lowering phase.

The data analysis included calculations of maximum amplitude and the mean of the rectified signals as shown in figures 3 and 4 and the root mean square (RMS) of the signals as shown in figures 5 and 6 throughout the entire biceps curl movement, including the descent, hold, and ascent. Notably, the data for the triceps of participant 1 did not show a similar trend to the biceps for any analysis.

The RMS was calculated from the rectified EMG signal, which was obtained by taking the absolute value of the EMG signal. This process ensured that both positive and negative values of the signal contributed positively to the RMS calculation, providing a measure of the overall strength of the signal. The RMS value for the biceps is significantly higher than that of the triceps, which suggests that the biceps are predominantly responsible for generating force during a bicep curl.

One challenge faced during the experiment was noise being produced in the EMG signal, distorting the data and decreasing the accuracy of the recordings.

On one occasion, the sensor was not placed properly on the skin of the test subject, and this produced a noisy signal. This issue was rectified by replacing the sensor with one that had a better adhesive. On another occasion the sensor placed on one of the subjects was being moved constantly with each movement, causing the production of another noisy signal. The sensor was then moved to a more viable position on the arm of the test subject. This eliminated most of the noise produced.

Other external sources that can introduce interference and lead to noisy EMG signals include power lines or electronic devices, and muscle fatigue from prolonged contractions, which can be mitigated by including rest periods in the experiment.

The Fast Fourier Transform (FFT) analysis obtained from the EMG signals during the bicep curls as shown in figures 7 and 8 show distinct patterns for the biceps and triceps muscles of subject 1 while curling the 6kg weight. The biceps graph shows a dominant peak at lower frequencies, reflecting synchronized muscle activation during the lifting phase, while the triceps graph displays a broader distribution of power across frequencies due to its role in the lowering phase. The biceps signal has a more defined peak at lower frequencies, indicating higher activation compared to the triceps. For further investigation, is would be advisable to increase the sample size by eliminating repetitions per weight per participant and instead increase the number of subjects performing bicep curls with varying weights. This would yield more data, improving the identification of trends while minimising participant fatigue. Additionally, conducting a t-test can assess the hypothesis by determining if there is a significant difference in the data, thereby guiding further analysis.

CONCLUSION

The aim of this experiment was to investigate the difference in muscle activation patterns between the triceps and biceps during bicep curls and to understand how various factors such as weight and gender influence these patterns.
An experiment was conducted using surface EMGs, several weights and two subjects, the data was recorded and then analysed using MATLAB.
The results consistently demonstrated higher muscle activation in the biceps compared to the triceps during throughout the experiment. This aligns with the anatomical function of the biceps as the primary flexor of the elbow joint and highlights its dominant role in generating force during the curling motion [5].
The biceps exhibited higher maximum amplitude and mean rectified value throughout the exercise. This indicates greater muscle activation in the biceps compared to the triceps, further confirming the difference in the roles of these muscles during the bicep curl movement.
Challenges were encountered during the experiment, such as noise interference in the EMG signals, which could have been addressed through better electrode placement and removing of external sources of interference, to ensure the accuracy of the data collected. Future experiments incorporating a larger sample size, better electrode protocols and structures put in place to reduce signal disturbance could increase the accuracy and reliability of this experiment.

REFERENCE LIST

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